

MAESTRO INSTANCE SETUP AT GOOGLE CLOUD PLATFORM

This document provides the information on the configuration of an instance in GCP Cloud for on premise installation of Maestro.

The described setup is necessary to enable necessary access and performance for Maestro application.

1.1 INSTANCE CONFIGURATION

- **Minimum instance type:** n4-standard-2 (2 vCPU, 8 GB RAM)
- **Disk size:** 100 GB
- **Operating System:** Ubuntu 22.04

1.2 INSTANCE SETUP FLOW

Before you install Maestro on the instance, it should be configured according to the following flow, using the **gcloud** tool:

1. Create a network where the Maestro resources will be deployed:

```
gcloud compute networks create maestro-onprem-network --subnet-
mode=custom
```

2. Create a subnetwork in a region of your choice:

```
gcloud compute networks subnets create maestro-onprem-subnet \
  --network maestro-onprem-network \
  --range 10.0.0.0/24 \
  --region europe-west1
```

3. Create a firewall rule that opens the ports necessary for Maestro work:

```
gcloud compute firewall-rules create maestro-onprem-firewall \
  --direction=INGRESS --network=maestro-onprem-network \
  --action=ALLOW --
rules=tcp:22,tcp:443,tcp:3000,tcp:8080,tcp:8200,tcp:9000,tcp:9001,tcp:
9090,tcp:15672,tcp:27017,icmp \
  --source-ranges=0.0.0.0/0
```

4. Create a static IP address to be assigned to Maestro instance:

```
gcloud compute addresses create "maestro-onprem-static-ip" --region europe-west1
```

5. Deploy the virtual instance where Maestro will be set up:

5.1. Generate an SSH key:

```
ssh-keygen -C ubuntu -f maestro-onprem-key -N ''
```

5.2. Deploy an instance:

```
gcloud compute instances create "maestro-onprem-vm" \
  --image-family ubuntu-2204-lts \
  --image-project ubuntu-os-cloud \
  --machine-type e2-medium \
  --zone europe-west1-c \
  --metadata=ssh-keys="ubuntu:$(cat maestro-onprem-key.pub)" \
  --network maestro-onprem-network \
  --subnet maestro-onprem-subnet \
  --address maestro-onprem-static-ip
```

6. Create a DNS zone to use for a DNS name on Maestro instance.

Provide a desired DNS name as the `--dns-name` parameter value. The example below creates a `gcp.maestro3.community` zone (you can use the name of your choice here):

```
gcloud dns managed-zones create maestro-onprem \
  --description="maestro-onprem-zone" \
  --dns-name=gcp.maestro3.community \
  --visibility=public
```

Note: To create a DNS zone, we need to have control over `maestro3.community` domain. This will allow us delegate subdomain `gcp.maestro3.community` to domain `maestro3.community`. To delegate the subdomain, copy NS records from the created `gcp.maestro3.community` zone:

☐	gcp.maestro3.community	NS	21600	Default
				ns-cloud-b1.googledomains.com.
				ns-cloud-b2.googledomains.com.
				ns-cloud-b3.googledomains.com.
				ns-cloud-b4.googledomains.com.

Then create an NS record in `maestro3.comminity` zone:

gcp.maestro3.community	NS	Simple	-	No	ns-cloud-b1.googledomains.com. ns-cloud-b2.googledomains.com. ns-cloud-b3.googledomains.com. ns-cloud-b4.googledomains.com.
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